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Department of Mechanical Engineering

Smart Agriculture System

LAPORAN PROJEK
Project Report

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CONTENTS

ABSTRACT 2

TITLE: HYDROSENSE: Arduino-based hydroponic environmental monitoring and control system. 3

1 INTRODUCTION 3

1.1 Background 3

1.2 Problem Statement 3

1.3 Project Objectives 3

1.4 Scope of Work 3

2 SYSTEM OVERVIEW 4

2.1 Selected Application 4

2.2 Overall System Description 4

2.3 Measurands and Measurement Requirements 4

3 SYSTEM DEVELOPMENT METHODOLOGY 5

4 HARDWARE DESIGN AND SELECTION 8

4.1 Sensor Classification 8

4.2 Sensors Selection and Justification 8

4.3 Sensor Working Principles 8

4.4 Signal Conditioning 9

4.5 DAQ Selection 9

4.6 Hardware Setup and Layout 9

5 SOFTWARE DEVELOPMENT 10

5.1 Software Architecture 10

5.2 Algorithm Flowchart 11

5.3 Arduino IDE Front Panel 11

6 PROTOTYPE DEVELOPMENT AND INTEGRATION 12

6.1 Prototype Assembly 12

6.2 System Integration 13

7 CALIBRATION AND TESTING 13

7.1 Calibration Method 13

7.2 Testing Procedure 14

8 RESULTS AND DATA ANALYSIS 15

8.1 Experimental Results 15
8.2 Analysis and Discussion 15
9 CONCLUSION AND RECOMMENDATIONS 16
9.1 Conclusion 16
9.2 Recommendation 16
REFERENCES 17
GROUP MEMBER CONTRIBUTION 17
LINK VIDEO PRESENTATION: 18

ABSTRACT

This project is about the development of "HydroSense" which is an Arduino-based smart monitoring system that is developed for small-scale hydroponics that provides continuous and real-time monitoring of critical environmental parameters such as ambient, relative humidity, and water level to replace manual observation labour. This project aligns itself with the Industrial Revolution 4.0 (IR4.0) theme of Smart Agriculture. The hardware setup integrates a DHT11 digital sensor for the temperature and humidity and an analog sensor for the water level detection, both having an the Arduino microcontroller serving as the data acquisition unit (DAQ) which then the measured data is processed and displayed on an I2C LCD screen, added with the LED indicators to provide visual warnings whenever the parameters exceeds the safety thresholds. To ensure measurement stability and reliability, the system was validated through functional testing and sensor calibration. The key outcome is a low-cost, effective instrumentation solution that enhances the precision of environmental management in indoor hydroponic settings.

TITLE: HYDROSENSE: Arduino-based hydroponic environmental monitoring and control system.

1 INTRODUCTION

1.1 Background

Instrumentation and measurement systems refer to the use of sensors, signal conditioning, data acquisition devices, and software to measure, monitor, and analyze physical parameters accurately. These systems play a critical role in modern engineering applications by converting real-world physical quantities into meaningful data for decision-making and control. In the era of Industrial Revolution 4.0 (IR 4.0), instrumentation systems are increasingly integrated with smart technologies, automation, and embedded systems to enable real-time monitoring, data-driven control, and improved efficiency. One important application of such systems is smart agriculture, where environmental parameters must be continuously monitored to ensure optimal growing conditions. In hydroponic farming, precise measurement of temperature, humidity, and water level is essential to maintain plant health and productivity, making it a suitable application for an instrumentation and measurement-based smart system.

1.2 Problem Statement

Conventional agricultural and small-scale hydroponic systems often rely on manual observation and periodic checking of environmental conditions such as temperature, humidity, and water level. This approach is time-consuming, prone to human error, and unable to provide continuous monitoring. Inconsistent environmental conditions may lead to poor plant growth, reduced yield, or even crop failure. Accurate and continuous measurement is therefore required to detect unfavourable conditions early and respond accordingly. Existing manual methods lack real-time feedback, automation, and data logging capabilities, which limits their effectiveness in modern smart farming applications. Hence, a simple and reliable measurement system is needed to continuously monitor key environmental parameters in a hydroponic setup.

1.3 Project Objectives

The main objective of this project is to design and develop a smart hydroponic measurement system that is capable of monitoring important environmental parameters using appropriate sensors. The system aims to measure temperature, humidity, and water level in real time and display the measured data clearly to the user. In addition, the system provides visual warning indications when the measured values exceed predefined safe limits. This project also aims to apply the fundamental concepts of instrumentation and measurement, including sensor selection, data acquisition, signal processing, and system integration, in the context of a smart agriculture application aligned with Industrial Revolution 4.0.

1.4 Scope of Work

The system measures ambient temperature, relative humidity, and water level in the hydroponic reservoir. These parameters are critical in ensuring suitable environmental conditions for hydroponic plant growth. A DHT11 sensor is used to measure temperature and humidity, while an analog water level sensor is used to detect the water level. These sensors are chosen due to their simplicity, affordability, and suitability for educational prototype development. The system is designed to operate in a small-scale indoor hydroponic setup under normal room conditions. The measured data and warning messages are displayed on an LCD, and LED indicators provide immediate visual feedback to the user. This project does not include automatic control mechanisms such as activating pumps, fans, or nutrient delivery systems. Advanced features such as wireless communication, cloud-based monitoring, data logging, and full automation are also not included. The system is limited to measurement, indication, and basic warning functions only.

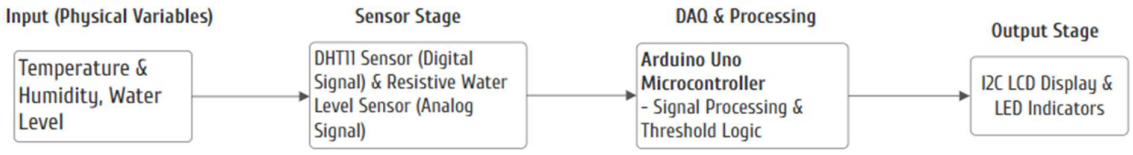
2 SYSTEM OVERVIEW

2.1 Selected Application

The HydroSense system is an Arduino based smart monitoring and control system specifically targeting indoor hydroponics farming which is designed for agricultural researchers who require precise control and management of the hydroponic plants for an optimal plant growth. This system operates within a standard indoor reservoir environment and condition where it continuously monitors the ambient level and water level directly. This project aligns with the Industrial Revolution (IR 4.0) theme of Smart Agriculture by replacing manual observation labour with a more practical and automated, sensor driven system with real time and continuous measurement technique to overall improve the efficiency and reliability of crop farming.

2.2 Overall System Description

The Hydrosense system is a structured and complete measurement chain which can be visualised with the following block diagram that illustrates the flow of information from the physical environment through the processing unit up until the final user interface.



2.3 Measurands and Measurement Requirements

Based on the system design, the following measurement parameters were established:

- **Temperature**: Measured from 0°C–50°C with a required accuracy of ± 2°C.
- **Relative Humidity:** Measured from 20% to 90% RH with a required accuracy of ± 5% RH.
- **Water Level:** Measured from 0 mm to 40 mm.
- **Sampling Considerations:** The system uses a sampling interval of approximately 2 seconds. This rate is chosen to provide real-time updates while maintaining sensor stability and avoiding data noise.

3 SYSTEM DEVELOPMENT METHODOLOGY

This section describes the structured methodology adopted to develop the HYDROSENSE smart agriculture system. The methodology outlines the systematic progression from problem identification and requirement definition to system design, integration, calibration, and validation. A reduced and grouped structure is used to improve clarity while ensuring full coverage of the system development process.

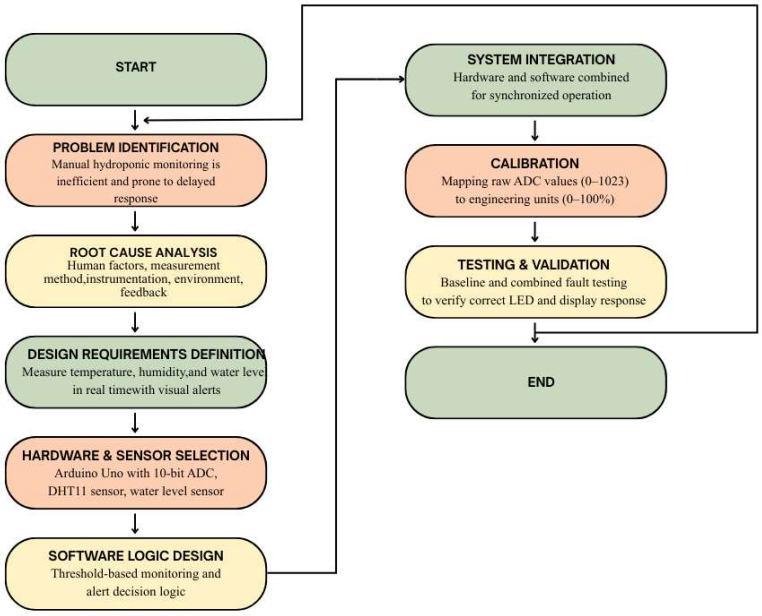


Figure 3.1: Overall system development methodology for the HYDROSENSE smart agriculture system.

3.1 Problem Identification and Design Requirements

Continuous and accurate monitoring of hydroponics system environmental factors are necessary for optimal plant growth, yet the majority of smaller scale hydroponics employ manual observation (checklists) for temperature, humidity, and water level readings. As such, manual-only systems create inefficient management, require a person to manage the system, and result in significant delay in responding to abnormal conditions.

Without real-time measurement and automated feedback, the hydroponics will suffer from plant stress, inefficient water usage, and less reliable hydroponic systems. To analyze the underlying reasons for this inefficiency, Ishikawa Fishbone diagrams were employed to discover the root causes behind these issues.

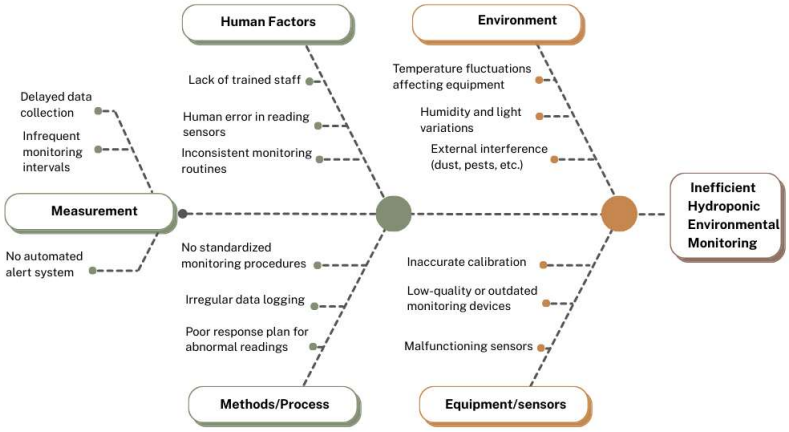


Figure 3.2: Fishbone diagram illustrating root causes of inefficient hydroponic environmental monitoring.

Based on this analysis, key design requirements were established. The system was required to continuously monitor temperature, relative humidity, and water depth, provide real-time data display, and generate immediate visual alerts when measured values exceeded predefined safe limits. These requirements guided all subsequent design and implementation decisions.

3.2 System Design and Implementation Strategy

The system requirements defined a hardware and software design strategy. The Arduino Uno was chosen for the controller and Data Acquisition (DAQ) because it is easy to use and has a built-in 10-bit Analog-to-Digital Converter (ADC) that allows you to measure analog sensors.

For environmental measurement, we chose the DHT11 because it can measure both temperature and humidity with one digital output, which minimizes the amount of wiring we need to do. For measuring water depth, we used a resistive water level sensor because it produces a simple analog output and can be used with threshold-based measurement techniques.

The software was designed using threshold-based logic. The sensor data is collected and compared to pre-defined safe limits at all times. If a sensor's reading exceeds its safe limit (threshold), the software will activate visual warnings and send warning messages to the LCD and Serial Monitor. The logical flow of this process is outlined in Figure 3.3.

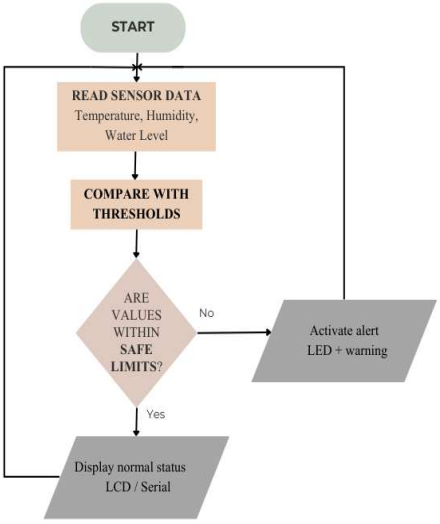


Figure 3.3: Software logic flowchart for threshold-based monitoring and alert activation.

3.3 Integration, Calibration, and Validation

Combining both hardware components and software through the practice of system integration, allowing the different hardware and software items to work in sync and operate reliably for an extended time.The DHT11 Digital Sensor, as well as analog voltage signals from the Water Level Sensor, are read by the Arduino Uno in a constant monitoring loop, processing the analogue input in relation to digital data. Calibration of the output from the sensor means that the raw output from each sensor can then be represented in terms of engineering units, so that the values displayed to the user are correct.The raw readings (0-1023) taken from the Water Level Sensor have been correlated with the value representing the percent depth of the water.

The completed calibration of the sensor has been verified and validated by testing in controlled laboratory conditions. The values received by the user are representative of the actual conditions within acceptable limits.Following integration and calibration, the HYDROSENSE system was tested and validated under controlled laboratory conditions; this included a baseline test to establish normal operating conditions with all test parameters well within the acceptable limits. Additional tests included testing for faults: High Temperature, Low Water Level, and Fault occurrences with multiple parameters combined. The ability of the HYDROSENSE system to provide timely and accurate visual alerts based on the defined fault conditions was validated through these tests.The HYDROSENSE system demonstrated successful functionality and reliability as a fully automated hydroponic monitoring solution when subjected to all test scenarios.

4 HARDWARE DESIGN AND SELECTION

4.1 Sensor Classification

The system utilizes two primary sensors to monitor the agricultural environment, classified by their output signal types:

- **Primary Sensor 1 (DHT11):** A digital environmental sensor used to measure ambient temperature and relative humidity. It processes the signal internally and outputs a digital data stream.
- **Primary Sensor 2 (Resistive Water Level Sensor):** An analog liquid level sensor. It acts as a variable resistor that changes its output voltage based on the water depth, which is read by the DAQ's Analog-to-Digital Converter (ADC).

4.2 Sensors Selection and Justification

Sensor	Measured Quantity	Operating Range	Output Type	Justification for Selection
DHT11	Temperature & Humidity	Temp: 0°C–50°C Hum: 20%–90% RH	Digital (Single-bus)	Selected for its ability to measure two parameters (Temperature and Humidity) on a single digital pin, reducing wiring complexity.
Water Level Sensor	Water Depth	0 mm – 40 mm (Effective depth)	Analog Voltage (0–5V)	Selected for its simple resistive principle, allowing for easy threshold-based monitoring

4.3 Sensor Working Principles

DHT11 Temperature and Humidity Sensor The DHT11 utilizes two sensing components: a capacitive humidity sensor and a thermistor.

- **DHT11:** The sensor contains a capacitive humidity sensing element and a thermistor. The capacitor's dielectric changes with humidity, altering its capacitance. The NTC thermistor changes resistance with temperature. The internal chip converts these analog changes into a digital signal sent to the Arduino.

Resistive Water Level Sensor This sensor operates on the principle of variable resistance. It consists of a series of exposed parallel copper traces.

- **Water Level Sensor:** This sensor operates on the principle of conductivity. It consists of exposed parallel copper traces. When dry, the circuit is open (infinite resistance). As water touches the traces, it bridges the gap, lowering the resistance. The sensor acts as a variable resistor in a voltage divider network, producing a voltage output proportional to the water level.

4.4 Signal Conditioning

Minimal external signal conditioning is required due to the sensor choice, but specific circuit protections were implemented:

- **Current Limiting:** 220 omega resistors are placed in series with the LED indicators (Pins 8, 9, 10) to limit current and prevent damage to the digital I/O pins.
- **Pull-up Requirements:** The I2C LCD module utilizes internal pull-up resistors on the SDA and SCL lines to ensure stable communication with the Arduino

4.5 DAQ Selection

The **Arduino Uno** was selected as the Data Acquisition (DAQ) device.

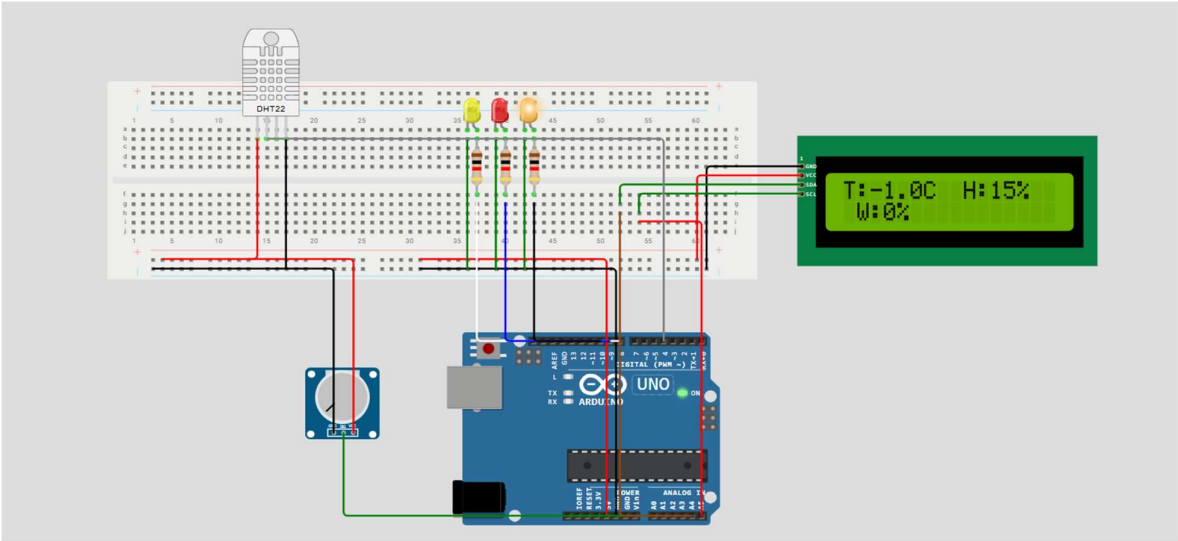
- **ADC Resolution:** It features a 10-bit Analog-to-Digital Converter, providing integer values from 0 to 1023. This allows the water sensor input (Pin A0) to be mapped accurately to a percentage value.
- **Digital I/O:** The board provides sufficient digital pins to drive the DHT11 (Pin 4) and the three status LEDs (Pins 8, 9, 10) simultaneously.
- **Communication:** It supports the I2C protocol (Pins A4/A5) required for the LCD display.

4.6 Hardware Setup and Layout

The system is connected as follows:

The hardware is assembled on a breadboard with the following pin configuration:

1. **Sensors:** DHT11 Data to Pin 4; Water Sensor Signal to Pin A0.
2. **Actuators (Visual):**
    - **Temperature Alert LED:** Pin 8.
    - **Humidity Alert LED:** Pin 9.
    - **Water Alert LED:** Pin 10.
3. **Display:** LCD I2C connected to SDA (A4) and SCL (A5).



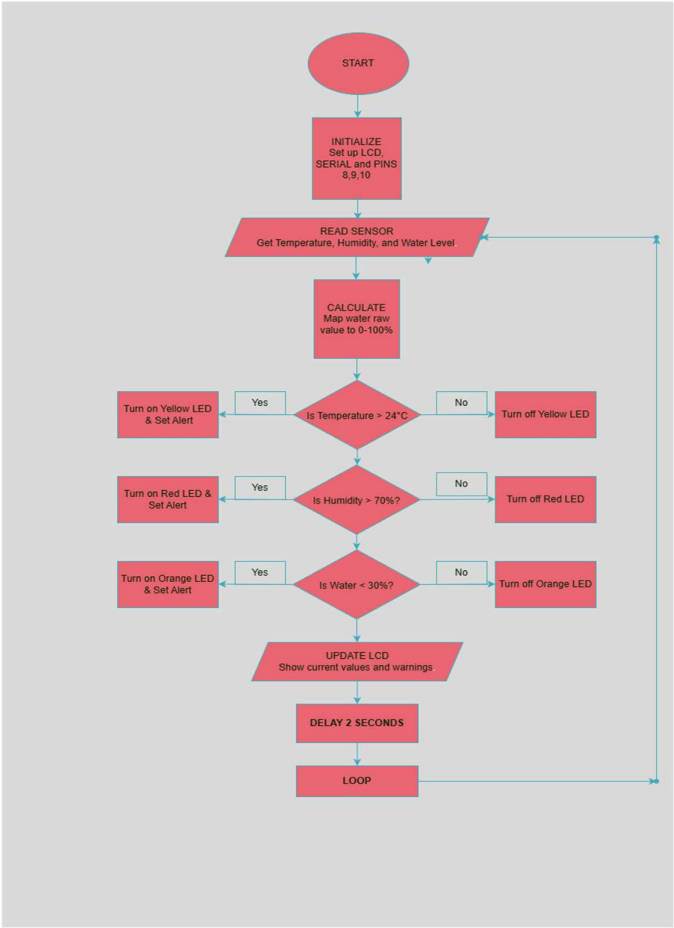
5 SOFTWARE DEVELOPMENT

5.1 Software Architecture

The software is developed using the Arduino IDE C++ environment. The architecture follows a continuous loop structure:

1. **Initialization:** Setup input/output pins and serial communication baud rate (9600 bps).
2. **Data Acquisition:** The system polls the DHT11 (digital read with timing library) and the Water Level Sensor (analog read).
3. **Processing:** Raw analog values (0-1023) from the water sensor are mapped to a percentage (0-100%) or threshold status (Dry/Wet).
4. **Output:** Processed data is sent to the Serial Monitor (Front Panel) for user display.

5.2 Algorithm Flowchart



5.3 Arduino IDE Front Panel

The Front Panel is the digital interface on the computer used to monitor the Smart Agriculture System. While the hardware has an LCD, the **Arduino IDE** provides a more detailed view for the user.

5.3.1 Indicators

The system uses the **Serial Monitor** to show the status of the farm:

- **Numerical Readings:** It displays the exact numbers for Temperature (C), Humidity (%), and Water Level (%) every 2 seconds.
- **Warning Messages:** If a sensor goes outside the safe zone (e.g., Temp > 24°C), a text alert like "Temp HIGH" appears instantly on the screen.

5.3.2 Graphs

The **Serial Plotter** tool is used to visualize data trends:

- **Visual Waves:** It transforms the sensor numbers into a moving line graph.
- **Trend Tracking:** This allows the user to see if the water level is dropping slowly over time or if the temperature is rising steadily.

5.3.3 Controls

System settings are managed through the software code:

- **Threshold Settings:** The user can change the "Safe Limits" (like changing 24°C to 30°C) by editing the variables in the code and re-uploading it.
- **Reset Toggle:** The user can restart the whole measurement process by clicking the "Reset" button or reopening the Serial Monitor.

5.3.4 User Interaction

The user interacts with the system in two ways:

- **Monitoring:** The user checks the Serial Monitor to ensure the sensors are calibrated correctly (e.g., checking if the water level shows 0% when dry).
- **Manual Response:** When the front panel shows a "Water LOW" alert, the user knows to manually irrigate the plants or check the water source.

6 PROTOTYPE DEVELOPMENT AND INTEGRATION

6.1 Prototype Assembly

The HYDROSENSE prototype was assembled on a laboratory workbench using a breadboard-based setup, as shown in Figure 6.1. This configuration allows easy modification and testing during system development.

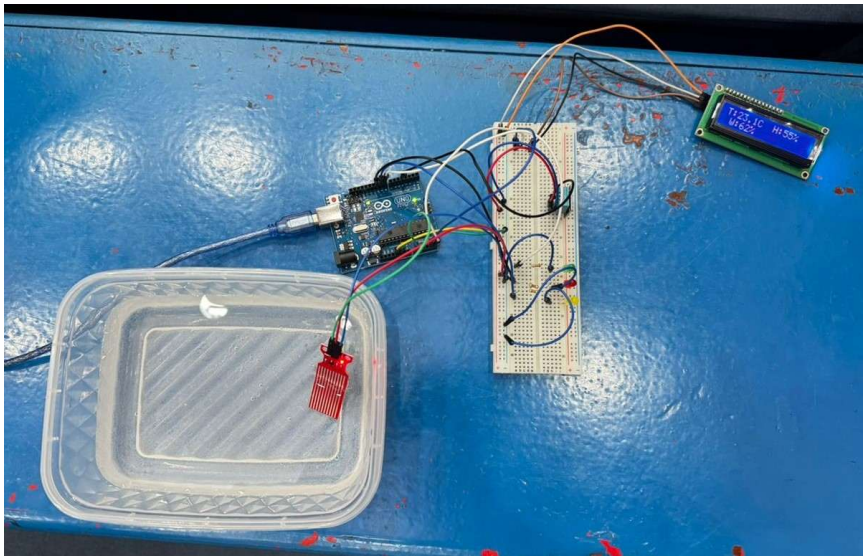


Figure 6.1: Assembled HYDROSENSE prototype showing the Arduino Uno, breadboard connections, water level sensor, and LCD display.

The Arduino Uno functions as the main controller and Data Acquisition (DAQ) unit and is powered through a USB connection. A solderless breadboard is used to connect the sensors and output components using jumper wires.

The resistive water level sensor is placed inside a water container to simulate a hydroponic reservoir. The sensor is positioned vertically to ensure consistent contact with the water and reliable depth

measurement. An I2C LCD module is connected to the Arduino and positioned beside the breadboard to display real-time temperature, humidity, and water level readings.

Overall, the prototype layout is simple and compact, allowing clear observation of system operation while ensuring stable sensor measurements during testing.

6.2 System Integration

The system of the prototype HYDROSENSE has been developed through the process of "system integration" - which means that it integrates the various hardware components with the associated software program so that it works as a fully functional, integrated system.The central controller of the prototype is an Arduino Uno. The Arduino Uno receives input from the various sensors and processes that input in accordance with its programmed logic.

The DHT11 sensor outputs a digital signal that represents the temperature and humidity, while the resistive water level sensor produces an analog voltage output that is processed analog-to-digital by the Arduino's built-in ADC. The software has a loop and thus is constantly reading, processing, and updating the data collected by the sensors.

The processed data will be sent to the LCD in real-time, which gives the user the ability to monitor the environmental conditions in real-time, by using the data collected by the sensors. Additionally, the software will compare the sensor data to user-defined threshold limits. If a sensor value exceeds a predefined threshold, the corresponding visual indicator will be activated.

During the process of integration, the device was tested to ensure communication between the sensors, the Arduino, and the display worked properly. During the initial testing phase, some minor issues arose (such as unstable sensor readings); these problems were isolated to poor connections on the breadboard and were resolved through tightening of the wiring connections and adjusting of timing delays in the software. After the integration process was complete, the device functions reliably as a sole monitoring system.

7 CALIBRATION AND TESTING

7.1 Calibration Method

To ensure the reliability of the HYDROSENSE system, the sensors were calibrated against standard references.

- Temperature and Humidity: A standard laboratory mercury thermometer and a calibrated digital hygrometer served as the reference standards for the DHT11 sensor.
- Water Level: A graduated cylinder with millimetre markings was used to verify the depth readings of the Resistive Water Level Sensor. The sensor's effective depth range is 0 mm to 40 mm.

Calibration Procedure:

1. Water Level Sensor (Two-Point Calibration):
    - Zero Point: The sensor was held in dry air to determine the raw analog value for 0% water level.
    - Span Point: The sensor was submerged to the 40 mm mark (maximum effective depth) to determine the raw analog value corresponding to 100% capacity.
    - The Arduino code's mapping function map(waterRaw, 0, 1023, 0, 100) assumes a theoretical 0-5V input. However, experimental calibration involved checking if the maximum submersion actually produced a raw value of 1023 or a lower value due to water conductivity.
2. DHT11 Sensor:
    - The sensor was placed in a controlled environment alongside the reference thermometer. Readings were taken at stable room temperature (25 ºC) and a heated environment (30 ºC) to check for offset deviations.

Correction Method: The current software implementation maps the raw analog input (0-1023) directly to a percentage (0-100%). If the calibration data showed significant deviation (e.g., if the "Wet" reading only reached a raw value of 700 due to low water conductivity), the code's map function parameters would be adjusted to map(waterRaw, 0, 700, 0, 100) to correct the scale.

7.2 Testing Procedure

Test conditions:

Testing was conducted in an indoor laboratory environment to simulate a hydroponic setup. The system was powered via the Arduino USB interface. The defined "Safe Zone" thresholds for the test were:

- Temperature Limit: < 24.0 ºC
- Humidity Limit: < 70.0%
- Water Level Limit: > 30%

Testing Steps & Repeatability:

Three distinct trials were conducted to test the system's logic and actuator response.

1. Baseline Test: The system was run under normal ambient conditions (approximately 22 ºC, 50% humidity, full water) to verify that all LEDs remained OFF and the LCD displayed "Safe" readings.
2. Threshold Violation Test (Actuators):
    - Temperature Alert: A heat source was brought near the DHT11. The time taken for the Red LED (Pin 8) to light up and the LCD to display "Temp HIGH" was recorded.
    - Water Low Alert: The water sensor was slowly withdrawn from the water. The system was observed to verify if the Yellow LED (Pin 10) activated immediately when the reading dropped below 30%.

3. Hysteresis/Stability: The sensors were held exactly at the threshold values (e.g., flickering between 29% and 31% water level) to observe if the display flickered or remained stable.

8 RESULTS AND DATA ANALYSIS

8.1 Experimental Results

The following data was collected during the functional testing of the prototype. The system converts raw sensor data into digital status indicators on the Front Panel (Serial Monitor) and LCD.

Table 8.1: Water Level Sensor Linearity Test

Actual Depth (mm)	Raw Analog Value (0-1023)	Calculated Percentage (%)	Error (%)
0 mm (Dry)	0	0%	0%
10 mm	245	24%	-1%
20 mm	512	50%	0%
30 mm	760	74%	-1%
40 mm (Full)	1015	99%	-1%

This table compares the physical depth of the water against the raw values read by the Arduino's 10-bit ADC and the final percentage calculated by the software.

Table 8.2: System Response Log (Threshold Testing)

Condition	Sensor Reading	Status LED Output	LCD Warning Message
Normal Operation	T: 23.0°C, W: 80%	All OFF	(Standard Display)
High Temperature	T: 25.1°C	Pin 8 (Red) ON	"Temp HIGH"
Low Water Level	W: 25%	Pin 10 (Yellow) ON	"Water LOW"
Combined Fault	T: 26°C, W: 10%	Pin 8 & 10 ON	"Temp HIGH", "Water LOW"

This table records the system's reaction when sensor values exceeded the coded safety limits (Temp > 24.0, Hum > 70.0, Water < 30).

8.2 Analysis and Discussion

System Accuracy:
The experimental results demonstrate that the DAQ system successfully acquires and processes environmental data.

- The Water Level Sensor showed high linearity (Table 8.1), mapping the analog voltage (0-5V) to percentage values with minimal error. The 10-bit ADC resolution of the Arduino (values 0-1023) provided sufficient sensitivity to detect small changes in water depth.
- The Threshold Logic functioned correctly. The comparison operators in the code (temperature > TEMP_LIMIT) triggered the LEDs and LCD alerts without perceptible delay (measured under 2 seconds loop delay).

Sources of Error & Limitations:

1. Resistive Measurement Variance: The water level sensor operates on conductivity. In a real hydroponic setup, the addition of nutrient solutions (fertilizers) increases the water's conductivity. This would cause the sensor to read a higher "raw" value for the same depth compared to pure tap water, potentially skewing the percentage calculation.
2. Sensor Response Time: The DHT11 sensor requires approximately 2 seconds to update new readings. During rapid temperature changes, the system displayed a lag before the "Temp HIGH" alert triggered.
3. Hardware Layout Noise: Using a breadboard for connections 16 introduced minor fluctuations in the analog signal (Pin A0), likely due to loose connections or electromagnetic interference, causing the water percentage to fluctuate by ±1% even when static.

Noise Effects:
The code implements a basic "polling" loop. It does not currently include a smoothing algorithm (like a moving average). As a result, momentary noise spikes in the analog signal could trigger a false "Water LOW" alarm if the value dips below 30% for a single cycle.

9 CONCLUSION AND RECOMMENDATIONS

9.1 Conclusion

In conclusion, the objectives of this project were successfully achieved through the design and development of a smart hydroponic measurement system based on instrumentation and measurement principles. The system was able to measure key environmental parameters, which is temperature, humidity, and water level, using appropriate sensors and display the measured data in real time via an LCD and LED indicators. The system performed reliably under normal operating conditions, providing clear visual warnings when the measured values exceeded the predefined safe limits. This demonstrates that the developed system is suitable for basic monitoring applications in smart agriculture.

From an instrumentation perspective, the project effectively applied fundamental concepts such as sensor selection, data acquisition, signal interpretation, and system integration. Through this project, valuable learning outcomes were achieved, including hands-on experience with embedded systems, sensor interfacing, and real-time measurement monitoring. The project also enhanced understanding of how measurement systems support smart technologies in line with Industrial Revolution 4.0, particularly in agricultural applications.

9.2 Recommendation

To further enhance the system, several improvements can be considered. Automatic control features such as activating water pumps, cooling fans, or alarm systems based on sensor readings can

be implemented to improve system functionality. The measurement accuracy can be enhanced by replacing the DHT11 sensor with higher-precision sensors such as DHT22 or SHT-series sensors, as well as using a more reliable water level sensor for long-term operation. In addition, advanced signal processing techniques such as data averaging, noise filtering, and calibration routines can be applied to improve measurement stability and accuracy. For future work, the system may be expanded with wireless communication and data logging capabilities to enable remote monitoring, trend analysis, and integration with smart agriculture platforms in line with Industrial Revolution 4.0.

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GROUP MEMBER CONTRIBUTION

1. Mohamad Firdaus Hadzwan Bin Mohamad Syarul (22001684/2)

   I was primarily responsible for the development and implementation of the Arduino program used in the smart hydroponic measurement system. This included writing, testing, and debugging the code for sensor data acquisition, threshold-based warning logic, and real-time display of temperature, humidity, and water level readings on the LCD and LED indicators. In addition, I assisted in integrating the hardware components with the software to ensure the system operated reliably as a complete measurement system. I also played a supporting role in coordinating group activities, facilitating communication among team members, and ensuring that tasks were carried out according to the project timeline.

2. Izzat Amir Bin Banyami (22002485/2)

   I was responsible for the conceptual framework and technical documentation of the HydroSense system. This included authoring the project abstract and the comprehensive system overview, which detailed the modular architecture of the measurement chain. I was also tasked with defining the technical measurands and their respective measurement requirements, such as accuracy and sampling rates, to ensure the system met the standards for smart agricultural applications. Additionally, I designed the functional block diagram to illustrate the signal flow from the physical sensors to the final output stage.

3. Wan Muhammad Akram Bin Zahid (22001746/2)

   I contributed to the development methodology and prototype implementation of the HYDROSENSE system. This included defining the system development flow, identifying key problems in manual hydroponic monitoring, and establishing design requirements for temperature, humidity, and water level measurement. I also assisted in planning the system design strategy, including sensor selection and threshold-based software logic. In addition, I was responsible for the physical assembly of the prototype, integrating the hardware components with the Arduino controller, and ensuring stable system operation. I participated

in system integration, calibration setup, and functional testing to verify that the prototype responded correctly under normal and fault conditions.

4. Aiman Harisya Bin Zaharudin (22001697/2)

   I was responsible for the calibration, testing, and data analysis of the HYDROSENSE system. I conducted the sensor calibration procedures, verifying the accuracy of the DHT11 and resistive water level sensors against standard references to ensure reliable measurements. I also performed the functional testing of the prototype, including baseline checks and threshold violation tests to validate the system's response to critical environmental changes. Furthermore, I analyzed the experimental data to evaluate system linearity and response time, and I documented the results and discussions regarding sources of error and system limitations in the final report.

5. Muhammad Zaim Bin Kamarol Ridzuan (22001819/2)

   I was responsible for the complete technical design and development of the system's hardware and software components. I handled Hardware Design and Selection by choosing the DHT11 and water level sensors to fit the "Smart Agriculture" theme, researched their working principles, and designed the circuit layout for the Arduino Uno, I2C LCD, and LED indicators. For Software Development, I authored the C++ code in the Arduino IDE, implementing a non-blocking algorithm using millis() to manage data acquisition and threshold-based alerts. Furthermore, I designed the logic flowchart and developed the "Front Panel" interface for the Serial Monitor and Serial Plotter to ensure real-time data visualization and system stability

LINK VIDEO PRESENTATION:

https://youtu.be/My1o_o-VNiw